

Unit 0

Review of Prerequisite Skills

Functions: Characteristics and Properties

Definitions:

Relation – a relationship between two (or more) things; typically represented by an ordered pair

Function – a relation in which every x -value (input) has **only one** y -value (output)

Domain – set of *input*-values

Range – set of *output*-values

Ex #1: Determine whether each relation is a function.

(a)

x	y
2	4
5	0
8	5
10	4

(b)

x	y
1	5
2	6
3	7
$\sqrt{9}$	8

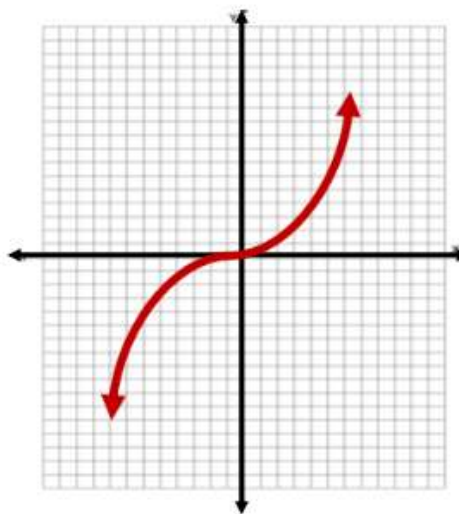
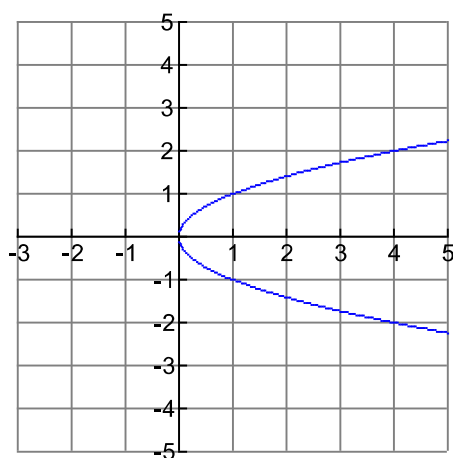
(c) $\{ (0,-5), (0,4), (2, 6), (3,7) \}$

(d) $\{ (1,-6), (2,6), (6, -6), (3,7) \}$

Vertical Line Test

f is a function if a vertical line can pass through the graph of $y = f(x)$ at most once.

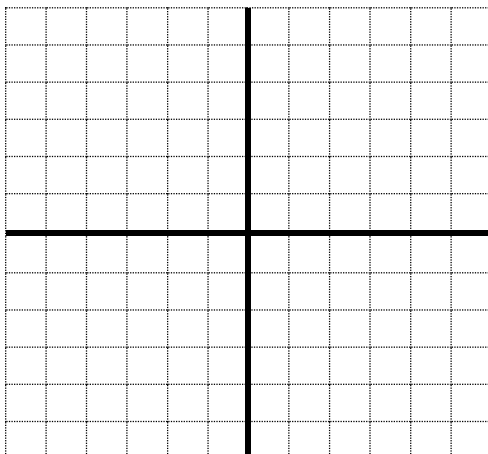
Alternately, if a vertical line passes through the graph of $y = f(x)$ two or more times, then f is not a function; f is only a relation.



Graph of Parent (basic) Functions

1. Linear $f(x) = x$

x	f(x)
-3	
-2	
-1	
0	
1	
2	
3	

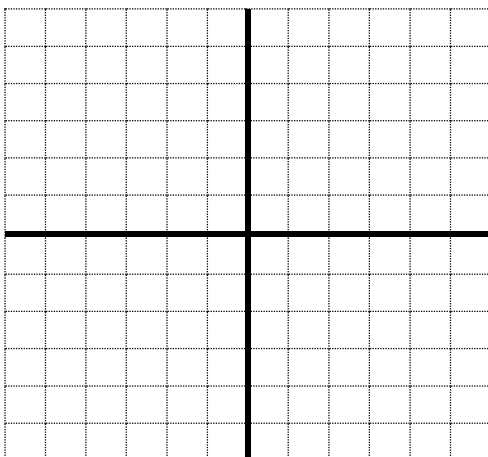


Domain_____

Range_____

2. Quadratic $f(x) = x^2$

x	f(x)
-3	
-2	
-1	
0	
1	
2	
3	



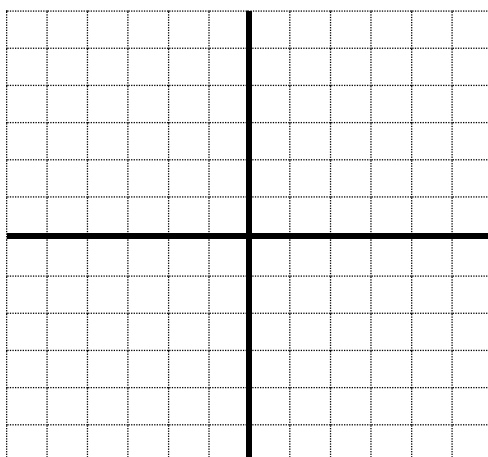
Domain_____

Range_____

Optimal Value_____

3. Cubic $f(x) = x^3$

x	f(x)
-3	
-2	
-1	
0	
1	
2	
3	



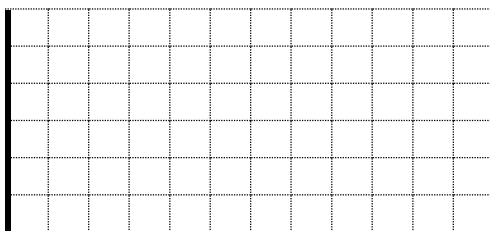
Domain_____

Range_____

Symmetry_____

4. Root $f(x) = \sqrt{x}$

x	f(x)
0	
1	
4	
9	

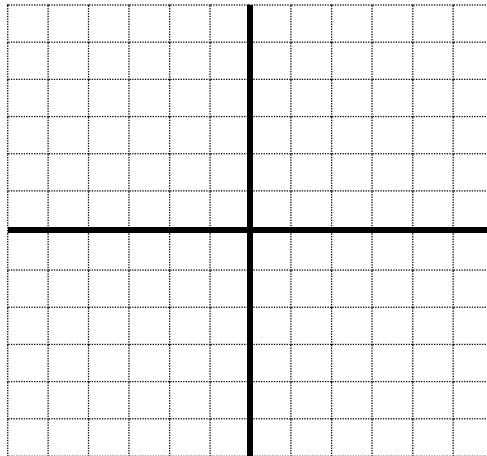


Domain_____

Range_____

5. Reciprocal $f(x) = \frac{1}{x}$

x	f(x)	x	f(x)
-3		3	
-2		2	
-1		1	
$-\frac{1}{2}$		$\frac{1}{2}$	
$-\frac{1}{3}$		$\frac{1}{3}$	



Domain_____

Range_____

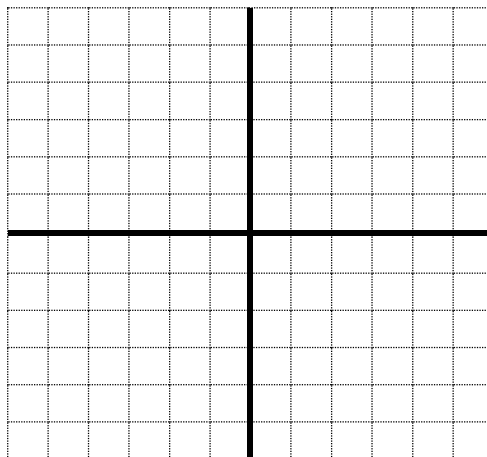
An **asymptote** is a line that a graph gets closer and closer to, but never actually touches. This graph has two asymptotes. What are they?

Symmetry_____

6. Absolute Value $f(x) = |x|$

The absolute value sign, $| |$, means to take the value of the number and drop the negative signs. For example, the absolute value of -5 which is written as $|-5|$ is 5.

x	f(x)
-3	
-2	
-1	
0	
1	
2	
3	



Domain_____

Range_____

Optimal Value_____

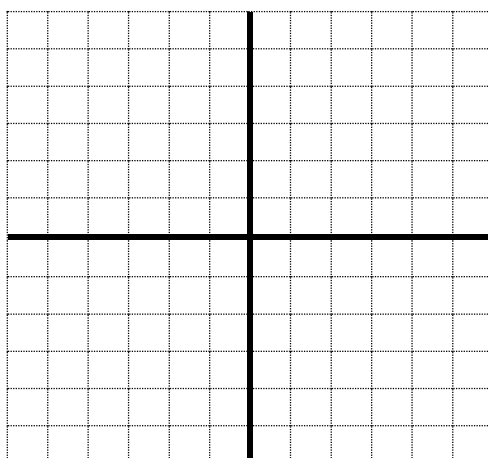
Domain over which function is decreasing_____

Domain over which function is increasing_____

Symmetry_____

7. Exponential $f(x) = 2^x$

x	f(x)
-3	
-2	
-1	
0	
1	
2	
3	



Domain_____

Range_____

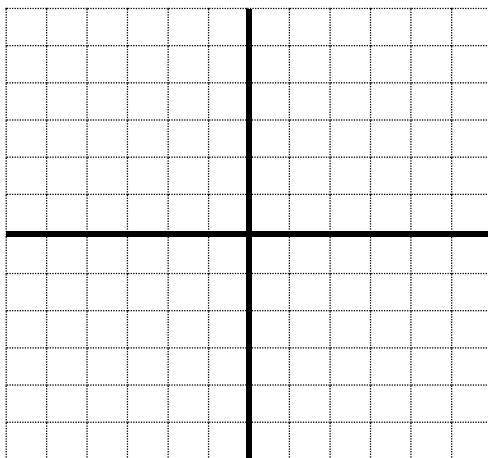
Y-intercept_____

This graph has one asymptote. What is it?_____

Increasing/decreasing?_____

8. Sinusoidal $f(x) = \sin(x)$

x	f(x)
-360	
-270	
-180	
-90	
0	
90	
180	
270	
360	



Domain _____

Range _____

y-intercept _____

x-intercept _____

Symmetry _____

Ex #2:

State the domain and range of each function.

(a) $y = -2(x - 5)^2 + 3$

$D =$ _____

$R =$ _____

(b) $f(x) = 3\sqrt{-2x+1} + 2$

$D =$ _____

$R =$ _____

(c) $f(x) = \frac{3}{x-1} + 2$

$D =$ _____

$R =$ _____

Inverse function

Definition:

A function is said to be **invertible** if the inverse of the function is also a function. Some functions ARE NOT invertible! The symbol $f^{-1}(x)$ is used for the inverse function and is pronounced, “ f inverse of x ”

One-to-one Function (1:1)

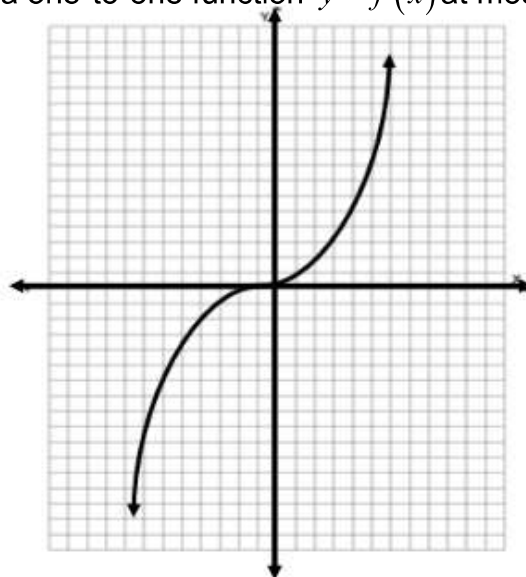
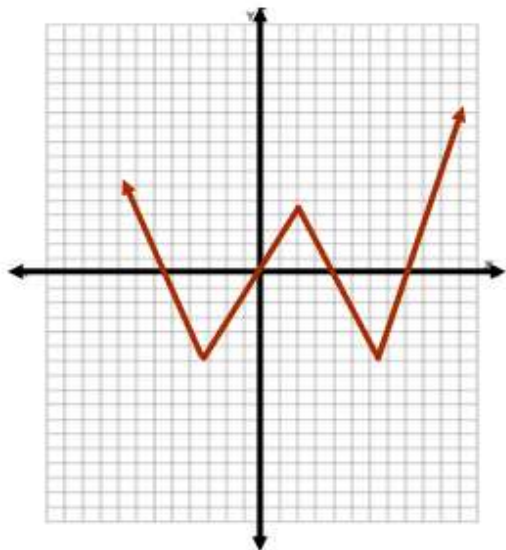
A function is one-to-one if its inverse is also a function. So, if a function is “invertible” then it must also be “one-to-one”.

One-to-one means that for each x there is exactly one y value and for each y value, there is exactly one x value.

One-to-one functions will pass the **Horizontal Line test**.

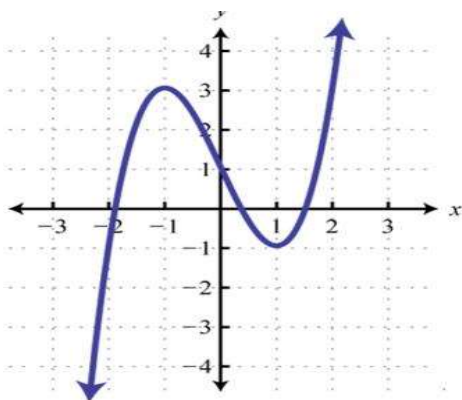
Horizontal Line Test

A horizontal line can pass through the graph of a one-to-one function $y = f(x)$ at most once.

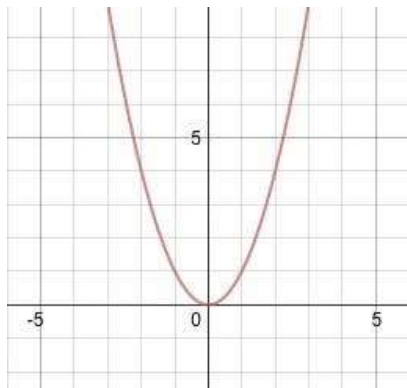


Ex#3: Which of the following functions is a one-to-one function?

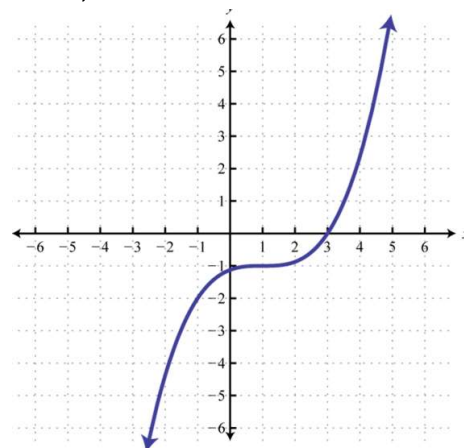
a)



b)



c)



How to find the inverse function?

- To find the inverse, switch the x and y variables, and then rearrange for y
- Switch the domain and the range
- Graphically it is the reflection in the line $y = x$

Ex #4:

Find the inverse function of the following functions:

a) $f(x) = -\frac{2x+1}{3x-2}$

b) $g(x) = \sqrt{2x+1}$

.

c) $k(x) = -2(x-3)^2 + 1$

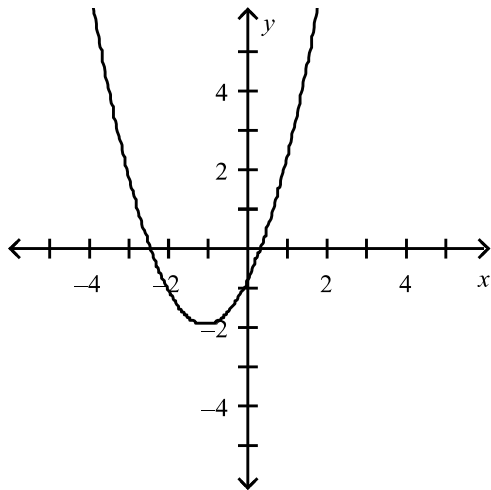
d) $h(x) = 2x^2 - 4x + 5$

Practice

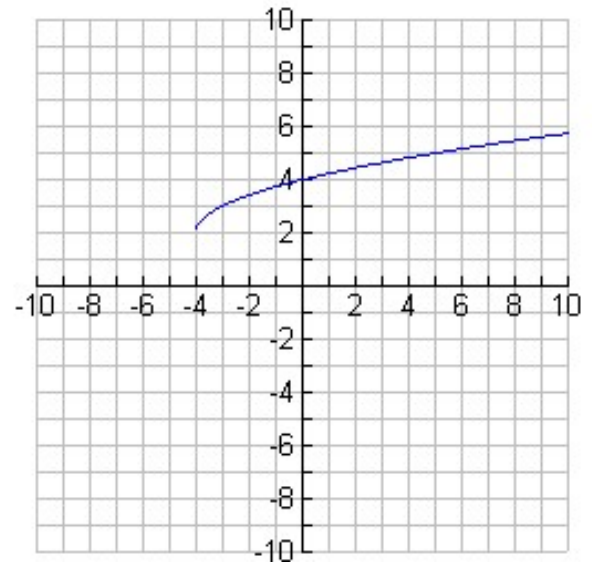
1. For each of the following:

- State the domain and range
- Draw the inverse (on the same grid) using colored pen or pencil
- State the domain and range of the inverse

i)



ii)



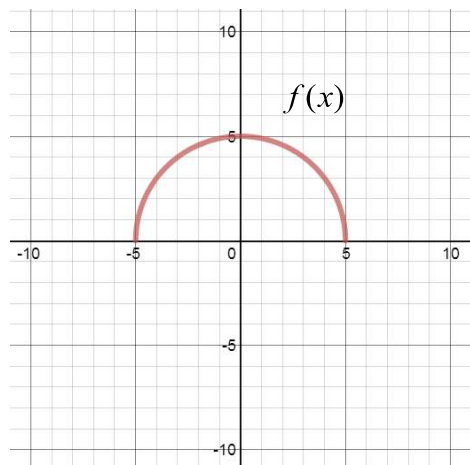
2. Given the two functions $f = \{(2,4), (3,1), (5,0), (6,-2), (-1,8)\}$ and $g(x) = -\sqrt{x-1} + 1$ determine:

- $f^{-1}(4)$
- $g^{-1}(x)$
- $f(g^{-1}(0))$

3. Given $f(x) = x^2 + 3x - 3$ and $g(x) = f^{-1}(x)$, what are the value(s) of $g(1)$?

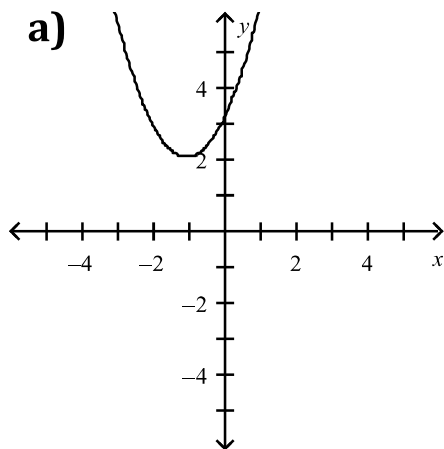
4. Given $f(x) = -k(4 + 2x)$, determine the value of k if $f^{-1}(2) = -5$.

5. Given the diagram below, graph $2f^{-1}(x+1)$ on the same set of axes;



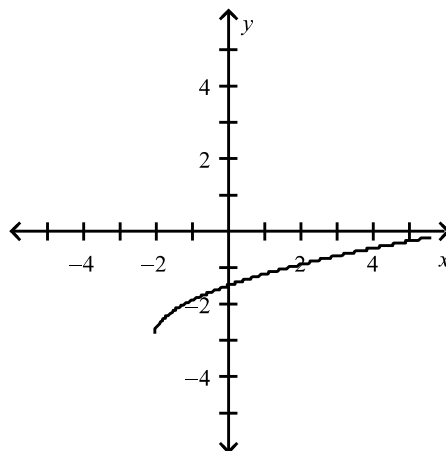
6. Find the equation of the following functions.

a)



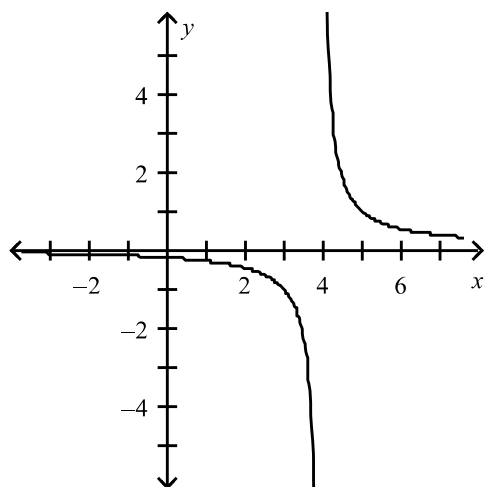
Equation: _____

b)



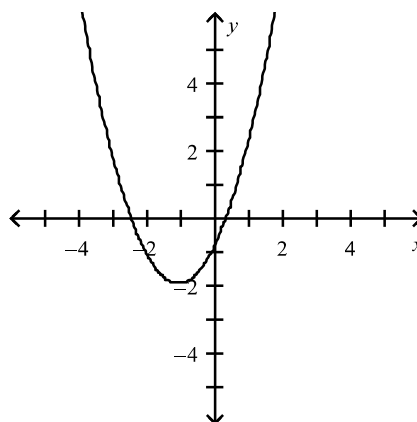
Equation: _____

c)



Equation: _____

d)



Equation: _____

Transformations of Functions

$|k| \geq 1$: Horizontal compression by a factor of $\left(\frac{1}{k}\right)$
 $0 < |k| < 1$: Horizontal stretch by a factor of $\left(\frac{1}{k}\right)$
 $k < 0$: Reflection in y - axis

$$y = a f(k(x - d)) + c$$

$|a| > 1$: Vertical stretch by a factor of (a)
 $0 < |a| < 1$: Vertical compression by a factor of (a)
 $a < 0$: Reflection in x - axis

$c > 0$: Translation c units up
 $c < 0$: Translation c units down

$d > 0$: Translation d units right
 $d < 0$: Translation d units left

Ex#1: For $f(x)=x^2$, describe $g(x) = -2f(3x + 6) - 1$

Ex#2: Find the equation of the transformed function as described below.

The graph of $y = g(x)$ has been reflected in the y-axis, vertically compress by a factor of one-third, horizontally stretched by a factor of two, shifted left three units. What is the equation of this transformed function?

Equation: _____

Using Method of Mapping Notation to Graph Transformed Functions

How to use mapping notation to graph transformed functions.

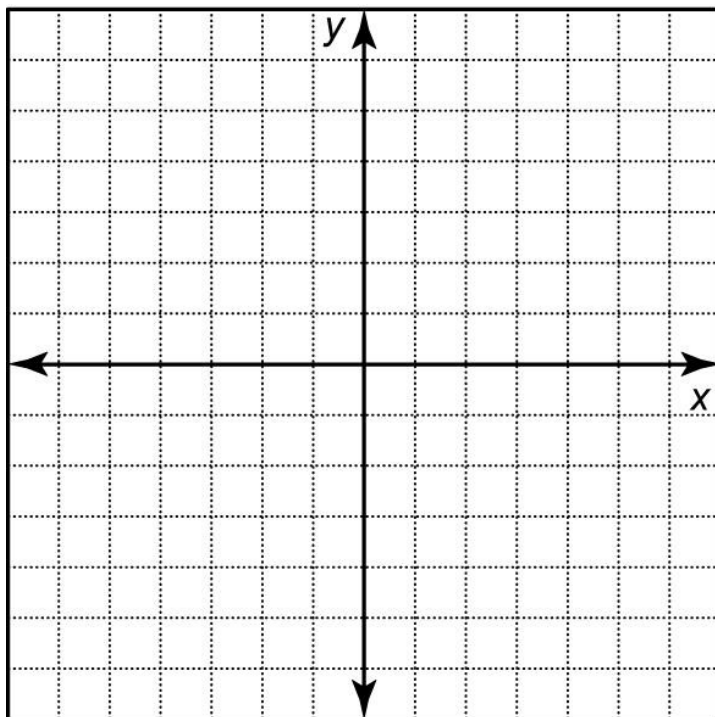
- 1) Create a table of values with original key points of base function
- 2) Create a table of values with NEW transformed coordinates
- 3) Plot original base functions and new functions

$$(x, y) \longrightarrow \left(\frac{1}{k}x + d, ay + c \right)$$

(x, y)	$\left(\frac{1}{k}x + d, ay + c \right)$

Ex #3 If $f(x) = x^2$ graph $g(x) = -3f[-2(x+1)] - 1$ using mapping rule.

(x, y)	



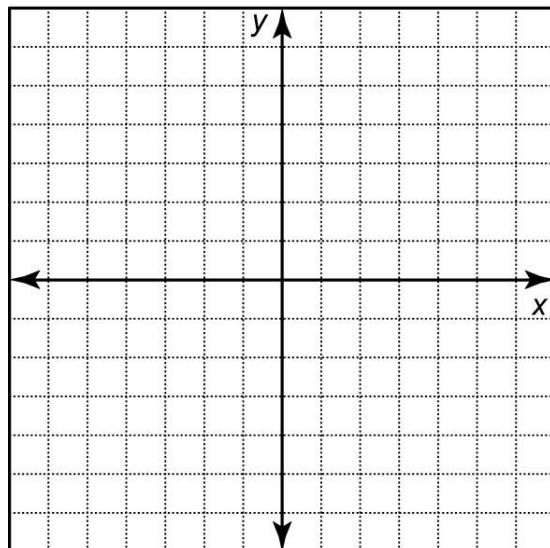
Practice

1. Graph the following functions by using mapping notation.

a) $f(x) = -3\sqrt{2(x-1)} + 4$

$(x, y) \rightarrow$

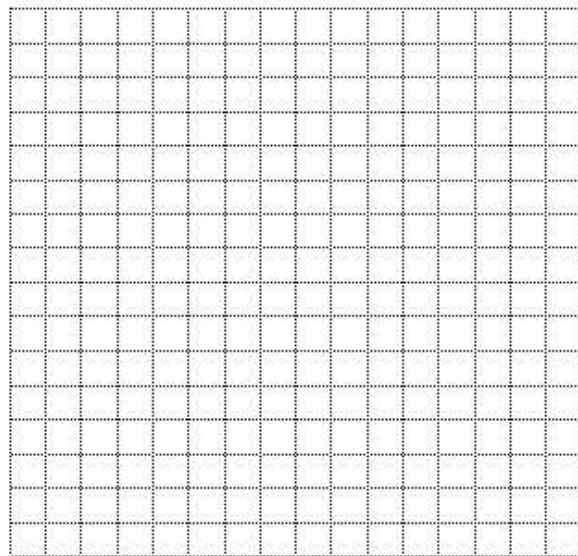
(x, y)	



b) $g(x) = -3\sin(3x - 45^\circ) + 4$

$(x, y) \rightarrow$

(x, y)	



2. If the point $(-10, -6)$ is on the graph $y = f(-2(x+9)) - 7$, find the original point.

Factoring

You should be very comfortable with the following types of factoring:

- **Common factoring**
- **Factoring trinomials**
- **Difference of squares**
- **Grouping**
- **Variable Replacement**

You can use these skills to factor trickier expressions!

a) $27x^5y^{-5} - 3x^3y^{-3}$

b) $\frac{1}{3}a^{-1}(a+4)^2 - 3a(a+4)^{-2}$

c) $x^2 - y^2 - 6x - 4y + 5$

d) $100y^2 - 4x^2 - 4x - 1$

e) $45x^2 - 210xy + 245y^2$

f) $2 - 20x^{-2} + 18x^{-4}$

g) $6(a+b)^2 + 17(a+b) + 5$

h) $3(x-y)^2 + 2x^2 - 2xy$

Practice

Factor fully.

Common Factor

a) $30x^5 + 9x^3 + 12x^2 - 6x^4$

b) $12(10x+7) - 3x(10x+7)$

Factor by Grouping

a) $5xy - 10x - 7y + 14$

b) $x^5 + 4x^4 + 2x^3 + 8x^2 - x - 4$

Trinomials

a) $2x^2 + 15x + 7$

b) $10x^2 + x - 3$

c) $x^2 - 8x + 15$

d) $x^2 + 7xy + 12y^2$

Difference of Squares

a) $9x^2 - 16$

b) $100x^6 - 49y^8$

Variable Replacement*

a) $(2x+1)^2 + 4(2x+1) - 12$

b) $6(x-5)^2 + 11(x-5) + 3$

c) $x^{10} - 7x^5 + 10$

d) $3x^{-4} + 31x^{-2} + 10$

e) $25(x+3)^2 - 36$

f) $100 - (2x+1)^2$

Grouping(Trinomials)

a) $x^2 + 6x + 9 - y^2$

b) $25 + 4x^2 - 9x^4 + 20x$

Common Factoring with Brackets

a) $3(x-3)^5 + 5x(x-3)^4$

b) $15x^4(2x+5)^3 + 18x^5(2x+5)^2$

c) $(2x+5)^7(3x-1)^4 + (2x+5)^6(3x-1)^5$

d) $3(3x-1)^4(x+3)^2 + 12(3x-1)^3(x+3)^3$

e) $4(2x+1)^{-3} + 5(2x+1)^{-4}$

f) $5x^2(3-2x)^{-5} - 10x(3-2x)^{-4}$

Set Builder, Real Number Line and Interval Notation

A **set** is a collection of unique elements.

Methods of Describing Sets:

Sets may be described in many ways: by roster, by set-builder notation, by interval notation, by graphing on a number line, and/or by Venn diagrams.

By roster: A **roster** is a list of the elements in a set, separated by commas and surrounded by French curly braces.

$\{2, 3, 4, 5, 6\}$	is a roster for the set of integers from 2 to 6, inclusive.
$\{1, 2, 3, 4, \dots\}$	is a roster for the set of positive integers. The three dots indicate that the numbers continue in the same pattern indefinitely. (Those three dots are called an ellipsis .)

By set-builder notation: **Set-builder notation** is mathematical shorthand for precisely stating all numbers of a specific set that possess a specific property.

$\mathbb{R}$ = real numbers; $\mathbb{Z}$ = integer numbers; $\mathbb{N}$ = natural numbers.

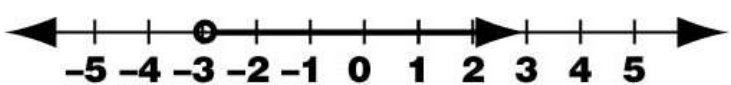
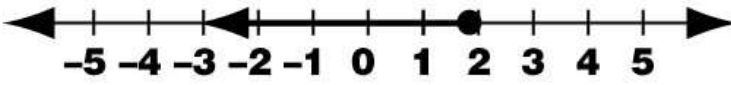
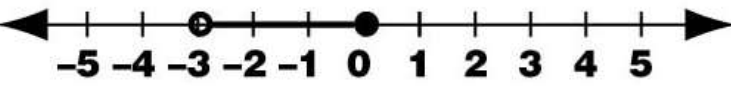
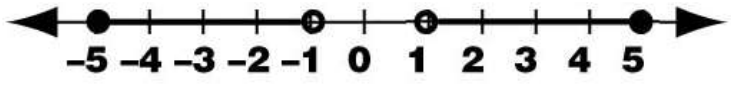
$\{x \in \mathbb{Z} \mid 2 \leq x \leq 6\}$	is set-builder notation for the set of integers from 2 to 6, inclusive. $\in$ = "is an element of" $\mathbb{Z}$ = the set of integers $\mid$ = the words "such that" The statement is read, "all x that are elements of the set of integers, such that, x is between 2 and 6 inclusive."
$\{x \in \mathbb{Z} \mid x > 0\}$	The statement is read, "all x that are elements of the set of integers, such that, the x values are greater than 0, positive." (The positive integers can also be indicated as the set $\mathbb{Z}^+$.)
It is also possible to use a colon (:), instead of the $\mid$, to represent the words "such that". $\{x \in \mathbb{Z} \mid 2 \leq x \leq 6\}$ is the same as $\{x \in \mathbb{Z} : 2 \leq x \leq 6\}$	

By the Real Number Line: It is standard to agree on the following conventions: To include an endpoint, we "bubble it in." To exclude an endpoint, we use an "empty bubble". Here is the set of all real numbers greater than -2 and less than or equal to



That is $\{x \in \mathbb{R} \mid -2 < x \leq 5\}$ or simply $-2 < x \leq 5$.

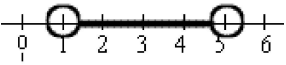
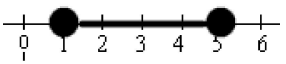
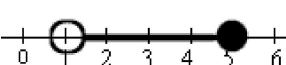
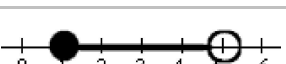
Ex #1: Graph the following intervals on a number line. Express each of the following graphs using set builder notation .

Number Line	Set builder
	
	
	
	

By interval notation: An **interval** is a connected subset of numbers. **Interval notation** is an alternative to expressing your answer as an inequality. Unless specified otherwise, we will be working with real numbers.

When using interval notation, the symbol:
[means "included" or "closed".
(means "not included" or "open".

Ex#2. The chart below will show you all of the possible ways of utilizing interval notation.

Interval Notation: (description)	Example	Interval Notation
Open Interval: (a, b) is interpreted as $a < x < b$ where the endpoints are NOT included.		$(1, 5)$
Closed Interval: $[a, b]$ is interpreted as $a \leq x \leq b$ where the endpoints are included.		
Half-Open Interval: $(a, b]$ is interpreted as $a < x \leq b$ where a is not included, but b is included.		
Half-Open Interval: $[a, b)$ is interpreted as $a \leq x < b$ where a is included, but b is not included.		
Non-ending Interval: (a, ∞) is interpreted as $x > a$ where a is not included and infinity is always expressed as being "open" (not included).		
Non-ending Interval: $(-\infty, b]$ is interpreted as $x \leq b$ where b is included and again, infinity is always expressed as being "open" (not included).		

For some intervals it is necessary to use combinations of interval notations to achieve the desired set of numbers.

As an inequality:	$x < -3$ or $x \geq 4$
In interval notation:	$(-\infty, -3) \cup [4, \infty)$
Notice that the word "or" has been replaced with the symbol "U", which stands for "union".	

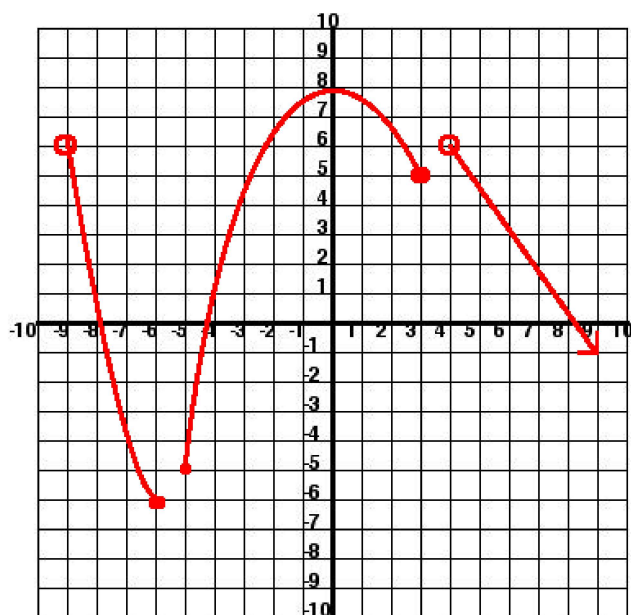
The following statements and symbols ALL represent the same interval:	
WORDS:	SYMBOLS:
"all numbers between positive one and positive five, including the one and the five."	$1 \leq x \leq 5$
"x is less than or equal to 5 and greater than or equal to 1"	$\{x \in \mathbb{R} \mid 1 \leq x \leq 5\}$
"x is between 1 and 5, inclusive"	$[1, 5]$

Ex#3. Complete the table:

#	Set Builder	Interval Notation	Number Line
1.	$\{x \in \mathbb{R} \mid x < -3 \text{ or } x \geq 3\}$		
2.		$(-4, 0) \cup (1, \infty)$	
3.			
4.	$\{x \in \mathbb{R} \mid -5 \leq x < 4\}$		
5.		$(-\infty, -2) \cup (2, \infty)$	
6.			
7.	$\{x \in \mathbb{R} \mid x < -3 \text{ or } -2 \leq x \leq 3\}$		
8.		$(-\infty, -4] \cup (-2, \infty)$	
9.			
10.	$\{x \in \mathbb{R} \mid x \leq -3 \text{ or } -1 < x < 4\}$		

Ex.#4 Determine the domain and range of the following relations using number line **and** interval notation.

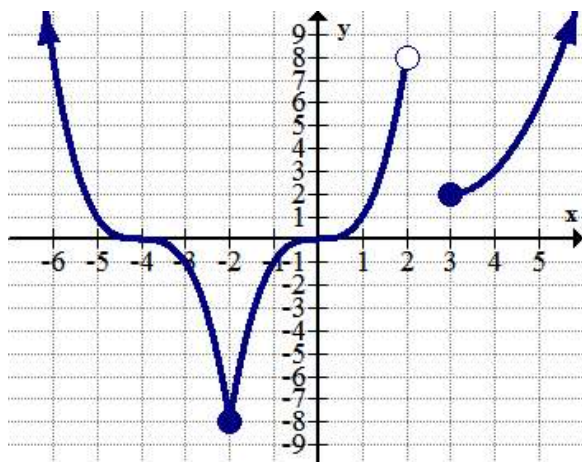
a)



Domain _____

Range _____

b)



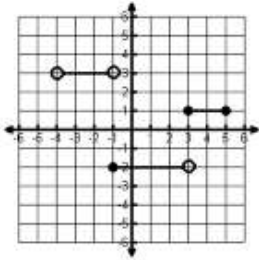
Domain _____

Range _____

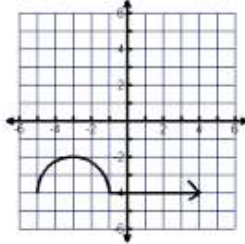
Practice

1. State the Domain and Range in Interval Notation. State if the graph is function or not.

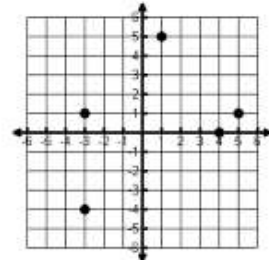
1) Domain _____
Range _____
Function? _____



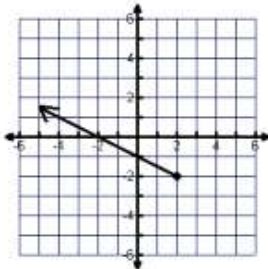
2) Domain _____
Range _____
Function? _____



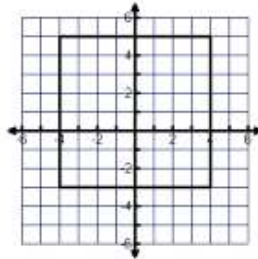
3) Domain _____
Range _____
Function? _____



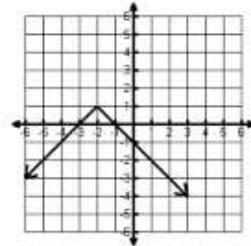
4) Domain _____
Range _____
Function? _____



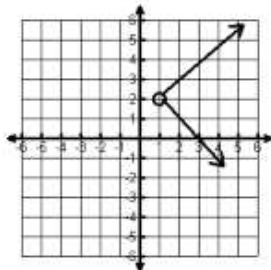
5) Domain _____
Range _____
Function? _____



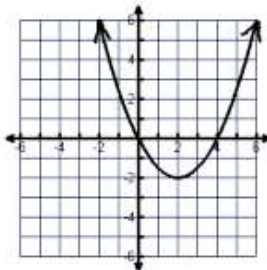
6) Domain _____
Range _____
Function? _____



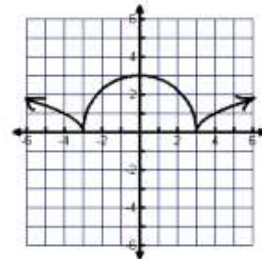
7) Domain _____
Range _____
Function? _____



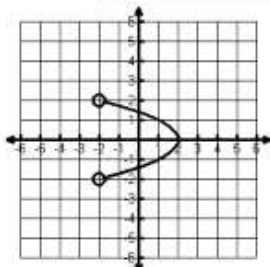
8) Domain _____
Range _____
Function? _____



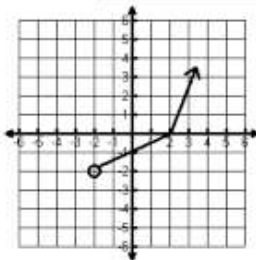
9) Domain _____
Range _____
Function? _____



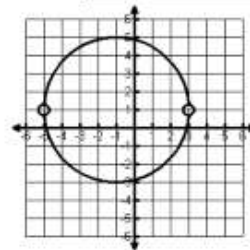
10) Domain _____
Range _____
Function? _____



11) Domain _____
Range _____
Function? _____



12) Domain _____
Range _____
Function? _____



2. Complete the table:

#	Graph	Set-Builder	Interval
1.		$x < -2$	
2.		$x \leq 4$	
3.		$x > 0$	
4.		$x \leq -6$ or $x \geq 3$	
5.		$x \leq 0$ or $x > 3$	
6.		$2 \leq x < 5$	
7.		$-7 \leq x < 0$ or $x \geq 4$	
8.			
9.			
10.			
11.			